

fn match_datetime_component

Michael Shoemate (@Shoeboxam)

This proof resides in “**contrib**” because it has not completed the vetting process.

Proves soundness of `match_datetime_component` in `mod.rs` at commit `f5bb719` (outdated¹).

`match_datetime_component` returns the data type and number of possible unique values of a compute operation that retrieves a datetime component.

1 Hoare Triple

Precondition

Compiler-verified

- Argument `temporal_function` of type `TemporalFunction`

Caller-verified

None

Pseudocode

```
1 def match_datetime_component(  
2     temporal_function: TemporalFunction,  
3 ) -> DataType | None:  
4     return ({  
5         Millennium: DataType.UInt32,  
6         Century: DataType.Int32,  
7         Year: DataType.Int32,  
8         IsoYear: DataType.Int32,  
9         Quarter: DataType.Int8,  
10        Month: DataType.Int8,  
11        Week: DataType.Int8,  
12        WeekDay: DataType.Int8,  
13        Day: DataType.Int8,  
14        OrdinalDay: DataType.Int16,  
15        Hour: DataType.Int8,  
16        Minute: DataType.Int8,  
17        Second: DataType.Int8,  
18        Millisecond: DataType.Int32,  
19        Microsecond: DataType.Int32,  
20        Nanosecond: DataType.Int32,  
21    }).get(temporal_function)
```

¹See new changes with `git diff f5bb719..4af7b91 rust/src/transformations/make_stable_expr/namespace_dt/expr_datetime_component/`
`rs`

Postcondition

Theorem 1.1. For any choice of `temporal_function`, returns the data type of the output if the input is an infallible row-by-row temporal expression, otherwise returns none.

Proof. For each choice of `temporal_function`, the data type is directly verified via testing. Infallible row-by-row is verified by inspection of the Polars source code. $\square$